

UET University Peshawar

Abbottabad Campus



Open Ended Lab

Lab Report No: 07

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Class No: 21

Course: SE-305L Computer Communications and Networks

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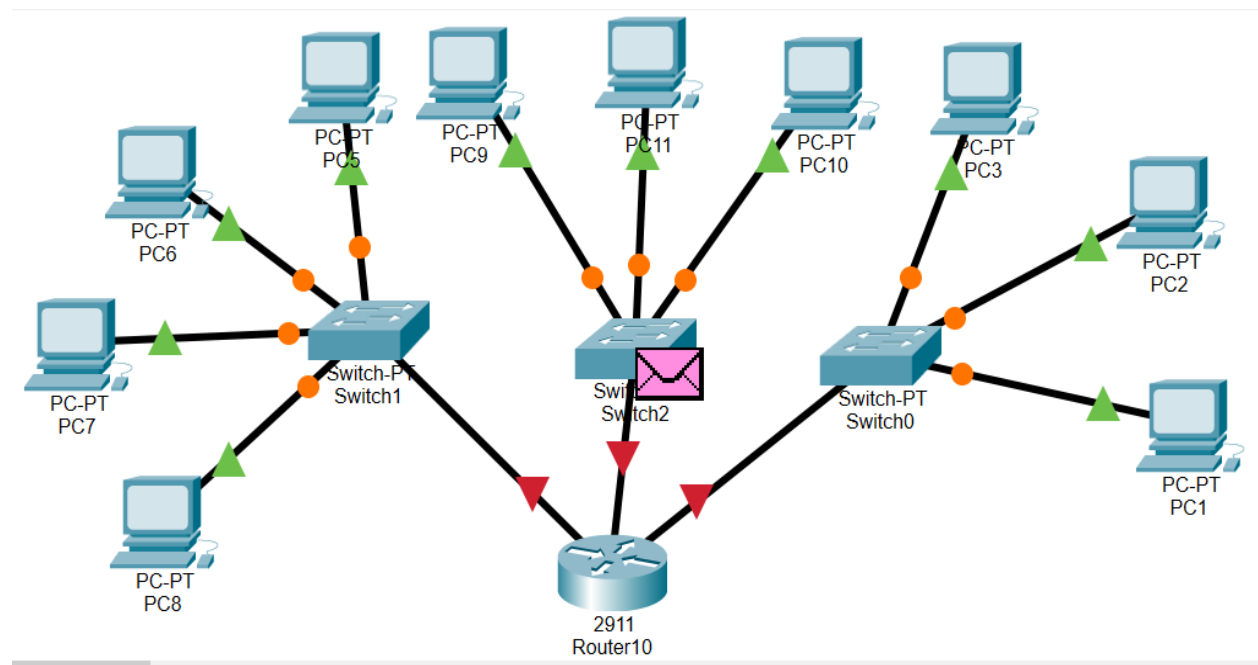
Lab 07: Designing a Corporate Office Network Using Packet Tracer.

Introduction

This lab focused on designing and implementing a scalable and efficient computer network. The primary goal was to understand device configuration, IP addressing, traffic simulation, and performance testing in a simulated environment using Cisco Packet Tracer.

We chose a **hybrid star topology** consisting of multiple switches connected to a central router. This design allows scalability, structured management, and easier troubleshooting.

Network Design



The topology contains:

- 3 switches (Switch0, Switch1, Switch2) forming the core.
- A central router (Router10) connecting all switches.
- 11 PCs distributed across the switches.

Devices Used

- **Router (Cisco 2911)**
Used for IP routing and DHCP services.
- **Switches (3x Cisco PT Switches)**
Provide LAN connectivity for PCs.

- **PCs (11x PC-PT)**
Represent clients for testing connectivity and simulations.

IP Addressing Scheme

- **IP Network:** 192.168.1.0/24
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.1.1 (assigned to router)
- **DHCP Range:** 192.168.1.2 - 192.168.1.100 (intended)

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip dhcp pool my_network
Router(dhcp-config)#my_network 192.168.1.0 255.255.255.0
                                     ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#NETWORK 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router (
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address
% Incomplete command.
Router(config)#ip dhcp pool my_network
Router(dhcp-config)#default-router(192.168.1.0)
                                     ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router (192.168.1.0)
                                     ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.1.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#lease 0 2 0
                                     ^
% Invalid input detected at '^' marker.

```

Simulating Traffic

Tests Performed

- **Ping Tests:**
 - PC to Router → **Success**
 - PC to PC (within and across switches) → **Success**
- **Packet Delivery:** Verified visually with green check marks and successful pings.
- **Email Simulation (Switch2):** Message sent between PCs across different switches → Successful.
- **DNS/HTTP/FTP:** Could be configured later with server roles on PCs.

Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	10.112	Switch0	PC2	CDP
	10.112	Switch0	PC1	CDP
	12.096	--	Switch2	STP
	12.097	Switch2	PC9	STP
	12.097	Switch2	PC10	STP
	12.097	Switch2	PC11	STP
	12.106	--	Switch1	STP
	12.107	Switch1	PC7	STP
	12.107	Switch1	PC8	STP
	12.107	Switch1	PC5	STP
	12.107	Switch1	PC6	STP
	12.111	--	Switch0	STP

Troubleshooting and Optimization

Issues Encountered

- **DHCP Pool Command Error:**
Incorrect CLI syntax initially caused DHCP setup to fail. This was resolved by re-entering correct commands.
- **Red Link (Router-Switch2):**
Initially inactive interface (no shutdown) – activated later.

Optimizations Made

- Excluded IPs from DHCP for static devices (e.g., router).
- Organized switches to reduce broadcast domains.
- Potential for VLANs to enhance traffic control.

Scalability and Performance

Adding More PCs

- 3 more PCs were added (total: 11).
- IPs successfully assigned dynamically using DHCP.

Performance Observations

- Ping response times remained under 10ms.
- No packet loss or routing issues.
- Switches handled increased load efficiently.